

ELITE MATHEMATICS

Pearson Edexcel International GCSE

# Modular Unit 1

## Adaptive Classified

OL-T20 · Completing the Square

### Questions

Student book · questions with writing space

■ 8 pages

# FAMILY MAP

## Completing the Square

**F01**

Complete the square for a non-monic quadratic

6 Q | 21 marks

**F02**

Complete the square for a monic quadratic

1 Q | 3 marks

**Working Unit 27**

4MA1-2H Q25

UNIT 1

3 marks

25 (a) Write  $28 + 24x - 6x^2$  in the form  $a - b(x - c)^2$  where  $a$ ,  $b$  and  $c$  are integers.

.....  
(3)

**Question 16**

4MA1-1H Q25

MODULAR UNIT 1

4 marks

**25**  $f(x) = 17 - 3x^2 + 12x$

Write  $f(x)$  in the form  $a - b(x - c)^2$  where  $a$ ,  $b$  and  $c$  are constants.

 $f(x) = \dots\dots\dots$

## 4MA1-2H Q19

3 marks

.....

**Question 15**

4MA1-1H Q24 Q24

MODULAR UNIT 1

4 marks

24 Express each of  $a$ ,  $b$  and  $c$  in terms of  $q$  so that

$$q + 12x - qx^2$$

can be written as  $a - b(x - c)^2$

 $a = \dots\dots\dots$  $b = \dots\dots\dots$  $c = \dots\dots\dots$

**Question 16**

4MA1-1H Q20 Q20(a), Q20(b)

MODULAR UNIT 1

4 marks

20 (a) Express  $7 + 12x - 3x^2$  in the form  $a + b(x + c)^2$  where  $a$ ,  $b$  and  $c$  are integers.

.....  
(3)

**C** is the curve with equation  $y = 7 + 12x - 3x^2$

The point **A** is the maximum point on **C**

(b) Use your answer to part (a) to write down the coordinates of **A**

(....., .....)  
(1)



21 Express  $5x^2 - 20x + 23$  in the form  $a(x - b)^2 + c$  where  $a$ ,  $b$  and  $c$  are integers.

(Total for Question 21 is 3 marks)